

Assignment 3

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Problem 1

P1 P2	Rock	Paper	Scissors	Probability
Rock	0,0	-1,1	1,-1	p1
Paper	1,-1	0,0	-1,1	p2
Scissors	-1,1	1,-1	0,0	1-p2
Probability	p1	p2	1-p1-p2	

No nash equilibria

Mixed Strategy Nash Equilibrium:

$$EU_1(R) = -p_2 + 1 - p - 1 - p_2 = -2p_2 - p_1 + 1$$

$$EU_1(P) = p_1 - 1 + p_1 + p_2 = p_2 + 2p_1 - 1$$

$$EU_1(S) = p_2 - p_1$$

$$EU_1(R) = EU_1(S) \implies -3p_2 = -1 \iff p_2 = \frac{1}{3}$$

$$EU_1(P) = EU_1(S) \implies 3p_1 = 1 \iff p_1 = \frac{1}{3}$$

Problem 2

P1 P2	P (Principal)	S (Secondary)	Probability
P	-1,-1	1,0	p
S	0,1	1,1	1-p
Probability	q	1-q	

Nash equilibria = (S,S)

Mixed Strategy:

$$\text{Player 1: } EU(P) = -q + 1 - q \quad EU(S) = 1 - q \quad EU(P) = EU(S) \iff q = 0$$

$$\text{Player 2: } EU(P) = -p \quad EU(S) = p + 1 - p \quad EU(P) = EU(S) \iff p = 0$$

$$q, p = 0 \implies (S,S)$$

Problem 3

Payoff function for Families:

$$U_i(k) = \alpha_1 * D_{home,k} + \alpha_2 * D_{work,k} * SP_i + \alpha_3 * SIB_{i,k}$$

Legend:

- $U_i(k)$ Payoff for family i if assigned to kindergarten k
- $D_{x,y}$ distance between x and y (Manhattan distance)
- SP_i 1 if single parent, 0 otherwise for family i
- $SIB_{i,k}$ 1 if child has sibling at kindergarten k, 0 otherwise for family i
- $\alpha_3 > \alpha_2 > \alpha_1$ weight parameters to reflect priority rules

Payoff function for Kindergartens:

$$V_k(M) = \sum_{i \in M_k} (\beta_1 * SIB_{i,k} + \beta_2 * SP_i * D_{work,k}) - \beta_3 * \max(0, |M_k| - C_k)$$

Legend:

- $V_j(M)$ Payoff for kindergarten k under matching M
- M_k set of families assigned to kindergarten k under matching M
- $SIB_{i,k}$ defined above
- SP_i defined above
- $D_{work,k}$ defined above
- $|M_j|$ is the number of children assigned to kindergarten k
- C_k capacity of kindergarten k
- $\beta_1, \beta_2, \beta_3$ weight parameters to reflect priority rules
- $\beta_3 \gg \beta_1$ and $\beta_3 \gg \beta_2$

In the case of twin sibilings (a,b) we can modify the payoff function for families so that if $k_a \neq k_b$ the function return $-\infty$.